

# CSE 566 Bütünleme Exam

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## I. QUESTION

Given two normal distributions  $p(x|C_1) \sim \mathcal{N}(\mu_1, \sigma_1^2)$ ,  $p(x|C_2) \sim \mathcal{N}(\mu_2, \sigma_2^2)$  and  $P(C_1)$ ,  $P(C_2)$ , calculate the Bayes' discriminant points analytically.

## II. QUESTION

For a two-class problem, for the four cases of Gaussian densities in Table 5.1, derive  $\log \frac{P(C_1|x)}{P(C_2|x)}$ .

## III. QUESTION

The following table shows the midterm and final exam grades obtained for students in a database course.

| Midterm Exam | Final Exam |
|--------------|------------|
| 72           | 84         |
| 50           | 63         |
| 81           | 77         |
| 74           | 78         |
| 94           | 90         |
| 86           | 75         |
| 59           | 49         |
| 83           | 79         |
| 65           | 77         |
| 33           | 52         |
| 88           | 74         |
| 81           | 90         |

- Use the method of least squares to find an equation for the prediction of a student's final exam grade based on the student's midterm grade in the course.
- Predict the final exam grade of a student who received an 86 on the midterm exam.

## IV. QUESTION

Let us say we have 2 states:

$S_1$ :  $\alpha$ ,  $S_2$ :  $\beta$  with equal initial probabilities. and the transition matrix is:

$$\mathbf{A} = \begin{vmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{vmatrix}$$

The emission probabilities for  $S_1$  is:

$$\mathbf{e}(S_1) = \begin{vmatrix} \text{A: } \frac{2}{5} \\ \text{G: } \frac{2}{5} \\ \text{T: } \frac{1}{10} \\ \text{C: } \frac{1}{10} \end{vmatrix}$$

The emission probabilities for  $S_2$  is:

$$\mathbf{e}(S_2) = \begin{vmatrix} \text{A: } \frac{1}{5} \\ \text{G: } \frac{1}{5} \\ \text{T: } \frac{3}{10} \\ \text{C: } \frac{1}{10} \end{vmatrix}$$

Decode the most likely sequence of states for sequence GGCT.

## V. QUESTION

Determine the number of different ways to partition a set of  $n$  elements into  $k$  clusters for the case, where clusters have names, clusters can be empty and elements have names.

## VI. QUESTION

Consider the alternative error function of one hidden layer multilayer perceptron for two-class discrimination problem:

$$E = - \sum_{t=1}^N (r^t \log y^t) ((1-r^t) \log(1-y^t)) + \alpha \sum_{i,j} w_{ij}^2 + \beta \sum_i v_i^2 \quad (1)$$

where  $N$  represents the number of instances,  $r^t$  represents the actual output,  $y^t$  represents the calculated output,  $w_{ij}$  are the weights between the inputs and the hidden units,  $v_i$  are the weights between the hidden units and the output unit. Derive the gradient descent update rules for the weights.